Below are 5 selected peer-reviewed and published manuscripts by Michael O'Donnell. For more recent work from the O'Donnell lab, please refer to "additional materials".

1. Alison Philbrook, **Michael P. O'Donnell**, Laura Grunenkovaite, and Piali Sengupta, Cilia structure and intraflagellar transport differentially regulate sensory response dynamics within and between C. elegans chemosensory neurons, *PLOS Biology* 22, e3002892 (2024)

   Summary: This study established that structural and transport components of primary cilia actively shape sensory neuron response dynamics. By linking ciliary architecture to neuronal coding properties, we demonstrated that subcellular structure directly influences how environmental stimuli are interpreted at the sensory level. Rather than treating cilia as passive sensory compartments, this work reframed them as active determinants of sensory computation. **My contribution:** I developed new signal analysis software used in all neural activity imaging and contributed conceptual input to the development of the project and the manuscript. **Intersection with ongoing work:** This work represents a mechanistic foundation for our independent interest in how cellular and metabolic state shapes sensory neural output. This led to our independent collaborative work with the Nechipurenko lab (WPI), studying *ric-8* in cilia biology (see "Additional Materials"), which extends these principles to understand how cell-specific ciliary regulation of GPCRs governs sensory function. Together, these studies move from structural determinants of signaling to understanding cell-specific modulation of GPCR function. This also provides foundational context for our research direction #2, in which we explore how GPCR signaling is modulated by microbial metabolites.

2. Munzareen Khan, Anna H. Hartmann, **Michael P. O'Donnell**, Madeline Piccione, Anjali Pandey, Pin-Hao Chao, Noelle D. Dwyer, Cornelia I. Bargmann, and Piali Sengupta, Context dependent reversal of odorant preference is driven by inversion of the response in a single sensory neuron type, *PLOS Biology* 20, e3001677 (2022)

   Summary: We demonstrated that context-dependent behavioral reversal can arise from polarity inversion within a single sensory neuron, revealing that modulation at the earliest stages of sensory processing is sufficient to reprogram behavioral valence. This study established a conceptual framework for understanding how internal state reshapes neural output without wholesale circuit rewiring. **My contribution:** I played a central role in conceptual development, directly co-mentoring both lead authors. I developed imaging and behavioral analysis software and co-wrote the paper. **Intersection with ongoing work:** This study directly relates to our lab's independent focus on how metabolic and microbial signals dynamically reshape sensory coding. This work conceptually links to our recent work on nuclear receptor ligand–neurotransmitter coupling, where we define the molecular logic by which metabolic state reconfigures sensory neuron function to alter behavior (2025 revision at *Nature*). I also use the datasets we generated in this paper as a key component of a coding bootcamp on behavioral analysis for incoming Interdepartmental Neuroscience Program PhD students (see "Teaching statement").

3. Chester J. J. Wrobel, Jingfang Yu, Pedro R. Rodrigues, Andreas H. Ludewig, Brian J. Curtis, Sarah M. Cohen, Bennett W. Fox, **Michael P. O'Donnell**, Paul W. Sternberg, and Frank C. Schroeder, Combinatorial assembly of modular glucosides via carboxylesterases regulates *C. elegans* starvation survival, *Journal of the American Chemical Society* 143, 14676–14683 (2021).

   Summary: In this interdisciplinary study, we uncovered that modular assembly of glucosylated metabolites may encode physiological state. Importantly, we identified microbe-dependent glucosylated neurotransmitter metabolites, revealing a new chemical language connecting diet, microbes, and host physiology. **My contribution:** I generated microbial gene knockouts in wild bacteria and performed isotope-labeling experiments critical for identifying these novel metabolites. **Intersection with ongoing work:** This work marks a shift toward incorporation of host metabolic pathways in interorganismal chemical communication, now a central axis of my independent laboratory. This work directly led to our recent work in which we identify an intertissue neurotransmitter glucosylation and trafficking pathway that is important for feeding-state dependent behaviors (in revision at *Nature*, see "Additional Materials). Additionally, the evolutionary perspective extends the implications of this work beyond nematodes, situating microbially influenced monoamines within a broader evolutionary framework.

4. **Michael P. O'Donnell***, Bennett W. Fox, Pin-Hao Chao, Frank C. Schroeder, and Piali Sengupta, A neurotransmitter produced by gut bacteria modulates host sensory behaviour, *Nature* 583, 415–420 (2020).

   Summary: In this study we identified the first microbe-derived neurotransmitter shown to modulate host sensory behavior. We demonstrated that bacterially produced tyramine alters host neural circuit output via host conversion to octopamine. This work established a direct chemical mechanism for gut–brain signaling and defined a new paradigm for microbiota-driven neuromodulation. **My contribution:** I conceived and led this project and wrote this manuscript as co-corresponding author. This work represents a clear intellectual break from my postdoctoral training and established the independent research direction that defines my laboratory. **Intersection with ongoing work:** This foundational discovery directly links to our current work, which moves from identifying a single microbe-derived neuromodulator to defining a broader regulatory architecture linking nuclear receptors, metabolic state, and neurotransmitter signaling (see “Additional Materials”). It also connects conceptually to our collaborative work with the Musser lab (Yale) which situates monoamine signaling within a deep evolutionary context, extending the conceptual reach of microbe-derived monoamines beyond host–microbe interaction into ancient mechanisms of tissue coordination.

5. **Michael P. O’Donnell***, PinHao Chao, Jan E. Kammenga, and Piali Sengupta, Rictor/TORC2 mediates gut-to-brain signaling in the regulation of phenotypic plasticity in C. elegans, *PLoS Genetics* 14, e1007213 (2018)

   Summary: We identified intestinal TORC2 signaling as a mediator of gut-to-brain communication that regulates sensory-driven plasticity. This work provided one of the first defined molecular pathways linking metabolic state to neuronal gene expression and behavior. **My contribution:** I conceived and led the project as co-corresponding author. This paper marked the first articulation of my long-term research direction: defining molecular mechanisms of gut-to-brain communication. **Intersection with ongoing work:** This study forms the mechanistic precursor to Additional Material #1 by establishing the intestine as a signaling hub capable of reshaping neural function.